

HoloBoard: an Immersive Teaching Board System

Jiangtao Gong
gongjiangtao2@gmail.com
Lenovo Research
Beijing, China

Xiaobai Ji
Lenovo Research
Beijing, China

Siyu Zha
Tsinghua University
Beijing, China

Teng Han
Institute of Software Chinese
Academy of Sciences
Beijing, China

Jiannan Li
University of Toronto
Toronto, Canada

Qicheng Ding
Lenovo Research
Beijing, China

Liuxin Zhang
Lenovo Research
Beijing, China

Qianying Wang
Lenovo Research
Beijing, China

ABSTRACT

We present HoloBoard, a next generation teaching board based on large-format, semi-transparent, interactive display technology that supports immersive teaching and learning. It can present immersive media curriculum materials including large-format videos, interactive 3D objects, AR effects, etc. Based on a multi-stage mixed-methods teacher-centered design thinking approach and a series of user studies with 6-9 years old primary students, the concept design of HoloBoard was defined and refined. The results showed that HoloBoard can be deployed in the wild and bring more engagement and interactivity in the primary school classroom, promoting students' spontaneous exploration and active learning.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**.

KEYWORDS

teaching board, immersive learning, large-format display, hologram, mixed reality

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1 INTRODUCTION

Teaching Boards have been used as an essential carrier to present information and help teachers communicate to a large audience in classrooms since the early 18th century [6]. In recent years, teaching boards have evolved into overhead projectors and interactive whiteboards [8, 9]. The importance of teaching boards in

education has been widely discussed, especially in displaying curriculum materials and enhancing teaching performance [4].

Nonetheless, teaching boards are currently still limited in terms of capability to be combined with virtual or augmented reality and present information in an immersive manner. As regard to immersive display technology, it is already well-perceived by the public, leading to commercially successful applications like VR game/movie¹ and AR videos². Its role in education has also been paid increasing attention to, for instance, virtual experiment simulation [1, 3] and AR-based word learning[5, 10]. However, the drawbacks of Head Mounted Display (HMD) or mobile-based AR have limited the promotion and application of AR/VR in education environments, not to mention the visual and body discomfort wearing HMD, disadvantages in facilitating collaborations, possible colliding danger during multiplayer mode, the purchasing and maintenance expense of headsets for schools [7].

This design presents the concept of HoloBoard, a large-format semi-transparent two-sided interactive display system. HoloBoard aims to blur the boundary between the physical classroom and the digital learning content, immersively displaying content without requiring teachers and students to wear HMDs. In other words, HoloBoard is developed based on the large-format holographic display technology and provides unique characteristics like large-FOV, coupling with reality, and interactivity to allow for more teaching activities requiring space and interactions. With HoloBoard, teachers can display slides and do two-sided freehand drawing on a large-format semi-transparent screen. Teachers can also use AR roleplay to facilitate situational teaching and interact with students via animated avatars or figures that students may like.

2 THE DESIGN OF HOLOBOARD

2.1 Design Thinking Process

The HoloBoard is designed through a lengthy process of user research. We conducted design interviews with six professional teachers to establish an initial understanding of traditional black-board teaching activities. After having a preliminary understanding

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¹<https://www.beatsaber.com/>

²<https://digiday.com/media/tiktok-is-coming-after-snapchat-with-a-new-augmented-reality-ad-format>



Figure 1: Traditional classroom (left), classroom with HoloBoard (right)

of teachers' requirements and expectations, we expanded the scope of user research and surveyed 26 professional teachers to understand their actual needs. The interview transcripts were analyzed using the open coding method [2]. Afterward, all codes were transcribed on sticky notes and arranged based on the teaching process via affinity diagramming. Finally, we reached a consensus on the general process of teaching and refined the concept of HoloBoard based on previous grounded analysis[2].

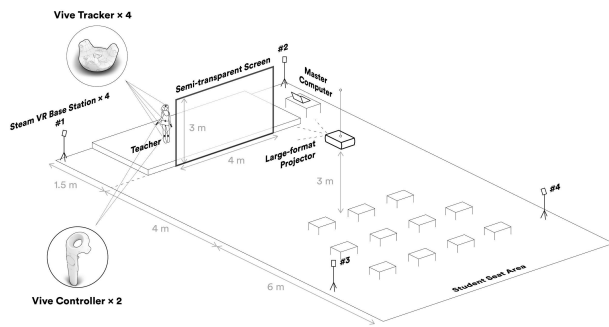


Figure 2: Hardware Composition of HoloBoard

2.2 Prototype Implementation

In terms of display, we used a 4m×3m large-format projected screen made of translucent (semi-transparent and semi-reflective) material for display in the front of the classroom. Behind the projected screen is the teacher's stage. To increase the immersion of the environment, a light-absorbing curtain is hung behind the stage absorbing redundant light passing through, and 4 adjustable spotlights are put above the staging area to cast more light on the teacher. Thus, the observer can see both the people behind the screen and the projected screen displays content at the same time.

4 SteamVR 2.0 base stations, and 4 Vive Tracker 2018 were used to implement motion tracking. We placed 4 base stations at the 4 corners of the classroom. The teacher does not need to wear a headset. By wearing 4 trackers on the head, waist, left and right feet, any movements within the range of the base stations can be tracked. The projector is an engineering sample with 7000 lm (later upgraded to 20000 lm). Considering mobility, we used Lenovo Y7000p (i7-9750H, NVIDIA RTX 2080 with Max-Q design, 32G RAM) as the control computer.

As regard to interactivity, we used 2 HTC Vive controllers to support interactions with digital content. The positions of the controllers are also tracked by the base stations.



Figure 3: Teacher wearing the trackers and carrying controllers. Note that the tracker on the teacher's head is not necessary.

2.3 Key Features of The Prototype

According to the design interviews and the surveys. We developed 4 key features for the working prototype.

Slides Control. The slides were made in Keynote/Microsoft Powerpoint and converted into a list of videos, which is later rendered in Unity. When the teacher clicks on the next/previous page button on the controller, it will automatically jump to the next/previous video in the video list.

Free-hand Drawing. In the teaching process, the teacher can start making ink annotations or free-hand drawing on the screen at any time by pressing and holding the 'Trigger' button and moving the controller. This function enhances the writing experience in traditional classrooms by combining annotations with interactive content.

Double-sided Augmented Reality. HoloBoard is double-sided symmetrical and delivers double-sided AR effect. Students can see the teacher's augmented performance such as role-playing, animations, special effects, etc., while the teacher can see the augmented student, such as the student's information displayed in the form of a dashboard floating above the student (two features cannot run at the same time).

Augmented Teacher with Roleplay/Animation. The spatial positions of the controllers and trackers worn on the teacher are mapped to a 2D/3D avatar model with animated skeleton structures. When the teacher moves, the 2D/3D avatar model will be rendered on the projected screen right in front of the teacher and follows the teacher's movement. From the students' view, the teacher is augmented by the avatars.

Augmented Student with Dashboard. When the teacher enters the Dashboard mode, each student's name tag and other learning-related information will be displayed above each student's head to provide teacher with more information. Furthermore, the teacher can give marks, complements and badges (like praise) to a certain student for encouragement. In this way, teachers can better understand the class in real-time and adjust the focus in the upcoming teaching activities.

VR Experiment. In this mode, the teacher can generate a 3D model of that learning content on the projected screen. In addition to this, by manipulating one or both controllers, the teacher can drag, scale, or rotate the 3D model and even make other VR experiments to illustrate in detail.

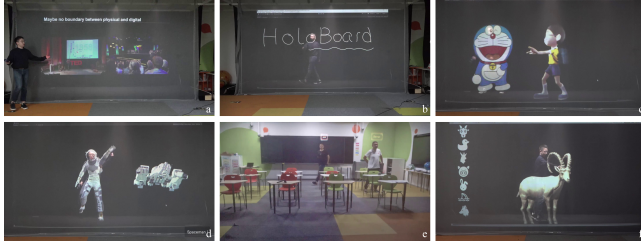


Figure 4: (a) Slides control, (b) Freehand drawing 'HoloBoard' on HoloBoard, AR roleplaying different characters by the teacher with 2D(c) and 3D(d) figures, (e) The teacher's view during the class. See the dashboard floating above two students, (f) User can explore objects in the 3D world with fully immersed virtual environment

3 EVALUATION

After the development of HoloBoard prototype, we conducted a round of evaluation. We collaborated with a STEM teacher (26 years old, female, master's degree in Education, 2 years of teaching experience) to design a STEM course for 6 to 9 year-old students introducing the features of the eight planets in the solar system. The lecture's content is based on the background of 'Space Exploration' and was about 25 minutes in length. We used this lecture to evaluate HoloBoard in real classroom settings and invited 15 student participants (ages 7.47 ± 0.884 , 7 male, 8 female). The evaluation showed that HoloBoard was conceptually new to both teachers and students, with the potentials to engage and interact with students effectively, promoting students' spontaneous exploration and active learning.

Most student participants were satisfied with the class content (4.27 ± 1.032). Higher scores were given on the form of this class (4.73 ± 0.704), showing that students were more satisfied with the form and the technologies we used during the study. Student participants reported that they do feel a sense of presence during interaction, and the role-playing session conducted by the teacher indeed merged the teacher herself into the scene and helped them learn gravity, temperature, and atmosphere of different planets. They felt the class was like a game, an experience they have never had before, and is very 'cool' compared to the traditional classroom environment.

4 DEMONSTRATION

Since our prototype is designed especially for educational situations, the experience of on-site demonstration without a typical classroom setting (chairs/desks/students) may not be complete. However, in our on-site demonstration, we expect participants to interact either with HoloBoard or with other participants by experiencing the 3D display, AR roleplay, and AR dashboard. After being introduced to



Figure 5: During the class, the teacher showed a Sun that literally made the kids wave their hands to cool down as if the Sun were present (a). The teacher also played an astronaut (d), viewed planets (f), viewed students' information via AR (e), and wrote on HoloBoard to annotate key takeaways (c).

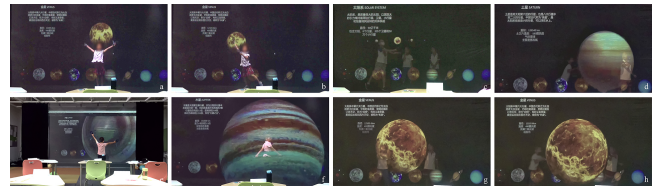


Figure 6: Afterclass students' spontaneous activities. (a)(b) A girl and (e)(f) a boy dance with a virtual planet in their hands like gymnastics. (c)(d) A boy is in control of the observing view while other students participate in and discuss different planet. (g)(h) Students collaborate to explore and have a better observation of each planet in the Solar System.

the system, each participant will be required to wear four trackers and invited to experience free-hand drawing, AR roleplay, and AR students (assisted by volunteers or other participants).

As to the online demonstration in response to the COVID-19, we plan on livestreaming to demonstrate several short but complete lessons featuring HoloBoard in a real classroom. The timetable of each class will be announced in advance and we do hope that it can be convenient for the audience from all the timezone to come and watch online. Taking place both in the morning and in the afternoon, classes will be covering different topics so that they will not bore people who plan on watching them all, but similar HoloBoard technologies will be demonstrated. Multiple cameras will be set up, providing several different angles of views for the audience's better observation. In addition, if the pandemic remains to be contained and travels in mainland China not banned, audience in China will be welcomed to physically come and not only experience but also audit the classes at the Lenovo HQ, Beijing, China where the HoloBoard prototype was built.

5 CONCLUSION

In this paper, we have presented the exploration of an immersive teaching board system, HoloBoard. By thoroughly investigating the application potential of a large-format semi-transparent two-sided interactive display in teaching and learning, we believe that this concept has the potentials for wider implementation, bringing

about more immersiveness, engagement, and interactivity in the primary school classroom.

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